Industrial AI (IAI) Final Project 2026

Project 4 – Aircraft Engine Remaining Useful Life Prediction

1. Background

At a large commercial airline operator, they are currently monitoring and collecting data from a fleet of aircraft engines. They have an extensive amount of historical data collected from their fleet, in which the historical records include data records from a new engine until its end of life. They would like to develop automated algorithms to assess the aircraft engine health and also predict how many remaining flights it can go before its end of performance life. Its end of life does not imply catastrophic failure of the engine but is more related to an unacceptable level of degradation and performance.

They currently have historical data from 260 aircraft engines. They are also currently collecting data from 100 different aircraft engines in the field. They collect a snapshot of the pressure, spool speed, temperature, and other parameters each flight. Each flight, the data could be collected in one of 6 different flight regimes (based on the altitude, Mach number, and throttle/power setting). From their experience and historical data, the high-pressure compressor degradation and efficiency was always the cause for retiring the aircraft engines. Thus their focus is to monitor the engine health with relation to the high pressure compressor (HPC) and predict the remaining life of the currently monitored engines with respect to the predicted HPC degradation level.

2. Problem Statement

Develop an automated health monitoring and remaining useful life prediction system for aircraft engines. A large amount of historical data from over 260 engines from their complete life cycle is provided. The objective is to use that historical data, to train health assessment and prediction models to accurately predict the remaining number of flights for the 100 aircraft engines currently being monitored (testing data set).

3. Sensor List and Example Plots

There are 24 different sensors used to monitor the aircraft engine. A listing of the 24 different sensors are provided in Table 1, and consist of various pressures, temperatures, rotational speeds, and operating settings of the engine. Each row in the data file represents one flight, and a snapshot of the 24 different sensor measurements are recorded at a particular time during the flight. There are 6 operating regimes in which the snapshot can be taken, and they are also labeled for each flight for each engine. For the training data, each engine starts out initially healthy and degrades over time until it is retired. An example degradation trend for one of the engines calculated using a regression based health method is shown in Figure 1. The engines typically have this same degradation pattern but degrade at different rates over time. The monitoring engines (testing data set) have some degradation over time but still have some remaining life until they have to be retired.

Table 1: List of Sensor Variables

VariableNumber	VariableName
1	Altitude
2	Mach #
3	Power Setting (TRA)
4	T2 Total temperature at fan inlet °R
5	T24 Total temperature at LPC outlet °R
6	T30 Total temperature at HPC outlet °R
7	T50 Total temperature at LPT outlet °R
8	P2 Pressure at fan inlet psia
9	P15 Total pressure in bypass-duct psia
10	P30 Total pressure at HPC outlet psia
11	Nf Physical fan speed rpm
12	Nc Physical core speed rpm
13	epr Engine pressure ratio (P50/P2) --
14	Ps30 Static pressure at HPC outlet psia
15	phi Ratio of fuel flow to Ps30 pps/psi
16	NRf Corrected fan speed rpm
17	NRc Corrected core speed rpm
18	BPR Bypass Ratio --
19	farB Burner fuel-air ratio --
20	htBleed (Bleed Enthalpy)
21	Nf_dmd Demanded fan speed rpm
22	PCNfR_dmd Demanded corrected fan speed rpm
23	W31 HPT coolant bleed lbm/s
24	W32 LPT coolant bleed lbm/s

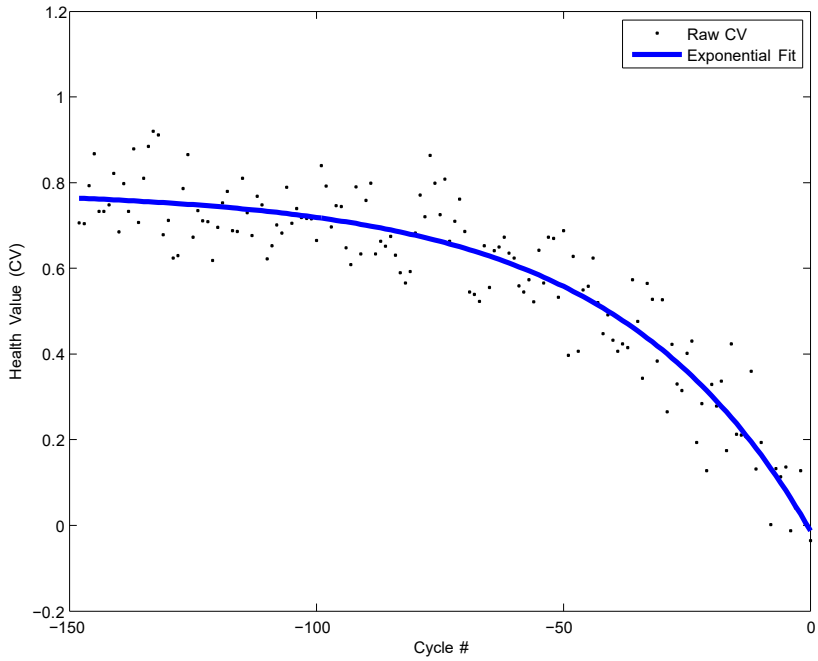


Figure 1: Example Degradation Trend from one engine (Using Regression Based Method)

4. Provided Data Set

a) Training Data Set

There are two structures in the provided data file, the first being "Train_EngineRun". This contains 260 engines inside the structure. In each structure, is the lifetime data for that engine, from a new engine until its retired. The data contains 24 columns, and each row corresponds to a given flight. For each flight, the data could have been collected in one of 6 regimes. A regime label for that flight is also provided. The "Variable_List" structure contains the sensor names that correspond to the 24 columns. The amount of rows (flights) varies for each engine, since some of the engines had shorter or longer life than others.

b) Testing Data Set

The testing data files are configured in the same way, but each engine unit only has some degradation and has additional flights left until it should be retired. The objective is to determine for each engine, how much additional life (# of flights) it has left. There are 100 engines in the testing data set, so there should be a life estimate for all 100 engines.

5. Suggested Algorithms

a) Feature Selection

a. Fischer Criterion

b) Health Assessment

a. Logistic Regression

b. Linear Regression

c. Self-Organizing Map - MQE

c) Remaining Useful Life Prediction

a. Trajectory Similarity Based Prediction

b. Neural Network Prediction

c. Regression / Curve Fitting

6. Submission of Test Results

The group will submit results for the test files (in excel or word or text file with an estimate of the remaining life (how many additional flights can it go) for each of the 100 engines in the testing data set.